

M.A. ECONOMICS THESIS · TRIBHUVAN UNIVERSITY

Land Fragmentation, Irrigation Access and Agricultural Production Sufficiency in Nepal

District-level evidence from the National Sample Census of Agriculture 2021/22

Sujil Devkota

Roll No. 075/72 · M.A. Economics

Patan Multiple Campus, Faculty of Humanities & Social Sciences

Supervisor: Dr. Kumar Bhattarai, Assistant Professor | 2026

The question in one line

Across Nepal, **55 out of every 100 farms** do not grow enough food for their own household for the year.

So the thesis tests two everyday explanations:

1. Broken-up land

Is it because farms are split into many small, scattered plots that are hard to work efficiently?

2. Missing water

Or is it because many farms depend on the monsoon and have no reliable irrigation?

BACKGROUND

Nepal's farming, in five numbers

All figures are from the National Sample Census of Agriculture 2021/22 and were checked against the official report.

4.1 M

agricultural holdings
(farms) in Nepal

0.54 ha

average farm size — about
one football field

2.8

separate parcels the
average farm is split into

55%

of farms cannot feed their
household from own
produce

54.5%

of farmland is irrigated —
very unevenly

Small farms, scattered plots, patchy irrigation, and widespread shortfall — the setting that motivates the study.

Three key terms, in plain words

Land fragmentation

One farm operated as several separate plots instead of one block.

Why it matters: scattered plots raise travel, supervision, and machinery costs, and make irrigation harder.

Measured as average parcels per farm.

Irrigation access

The share of a district's farmland that has a water supply beyond rainfall.

Why it matters: water lets farmers crop more than once a year and worry less about the monsoon.

Measured as irrigated-area share.

Production sufficiency

Whether a farm grows enough to feed its own household for the year.

Note: it is self-reported by the farmer — a measure of own-production vs. need, not of poverty.

The study models its opposite: insufficiency.

Objectives and hypotheses

Three research questions — how fragmentation and irrigation vary, how sufficiency varies, and how they relate — map onto three objectives and three testable hypotheses.

Objectives

1. Examine the district-level patterns of land fragmentation and irrigation access.
2. Examine the district-level patterns of agricultural production sufficiency.
3. Analyse the relationship between them — including whether irrigation moderates the fragmentation–sufficiency link.

Hypotheses

- H1 Wider irrigation coverage → lower insufficiency.
- H2 More fragmented holdings → higher insufficiency.
- H3 Irrigation moderates the fragmentation–insufficiency link (direction left open).

The data: one official census, every district

Source

National Sample Census of Agriculture 2021/22 — the government's official farm census, published by the National Statistics Office.

Coverage

All 77 districts of Nepal — the complete set, not a sample. One row of data per district.

What's in it

Parcels, irrigation, sufficiency, plus controls: farm size, loans, subsidy, machinery, market access, climate impact.

Trustworthy & reproducible

Built from published tables with documented scripts; every national figure was checked against the source report.

How the analysis works — three steps

1 Describe

Rankings, province profiles and maps show where each condition is high or low.

2 Correlate

Simple correlations check which conditions move with insufficiency.

3 Model

Regressions estimate each link while holding the other factors equal.

Two model ideas

Fractional logit — a regression built for an outcome that is a share between 0% and 100% (here, the insufficiency share), so predictions never fall below 0 or above 100.

Fixed effects & controls — a way of comparing 'like with like', so a link is not just an accident of which province or which farm size a district has.

The empirical model, precisely

Estimating equation

$$Y = \beta_0 + \beta_1 \cdot \text{Frag} + \beta_2 \cdot \text{Irrig} + \beta_3 \cdot (\text{Frag} \times \text{Irrig}) + \gamma \cdot \text{Controls} + \text{province FE} + \varepsilon$$

Y = the district agricultural-insufficiency share (one observation per district). Fragmentation and irrigation are mean-centred (2.81 parcels; 0.525 irrigated share), so the main effects read at the average holding.

$$H1: \beta_2 < 0 \quad H2: \beta_1 > 0 \quad H3: \beta_3 \neq 0$$

How it is estimated and tested

Two estimators. A weighted OLS benchmark (readable in percentage points) and the preferred fractional logit, by QML (Papke–Wooldridge), which keeps predictions inside [0,1].

Robust HC1 standard errors. Districts weighted by total holdings, so the estimand is the average holding, not the average district.

Average partial effects convert the logit coefficients to percentage points; a 999-replication pairs bootstrap gives distribution-free confidence intervals.

77 districts, 16 parameters. A parsimonious 5-term specification is estimated alongside; conclusions rest on the stability of signs across specifications, not on any single p-value.

Finding 1 — Nepal is not one story

Every condition varies enormously across districts:

Insufficiency share

32% (Rupandehi) to 87% (Dolpa)

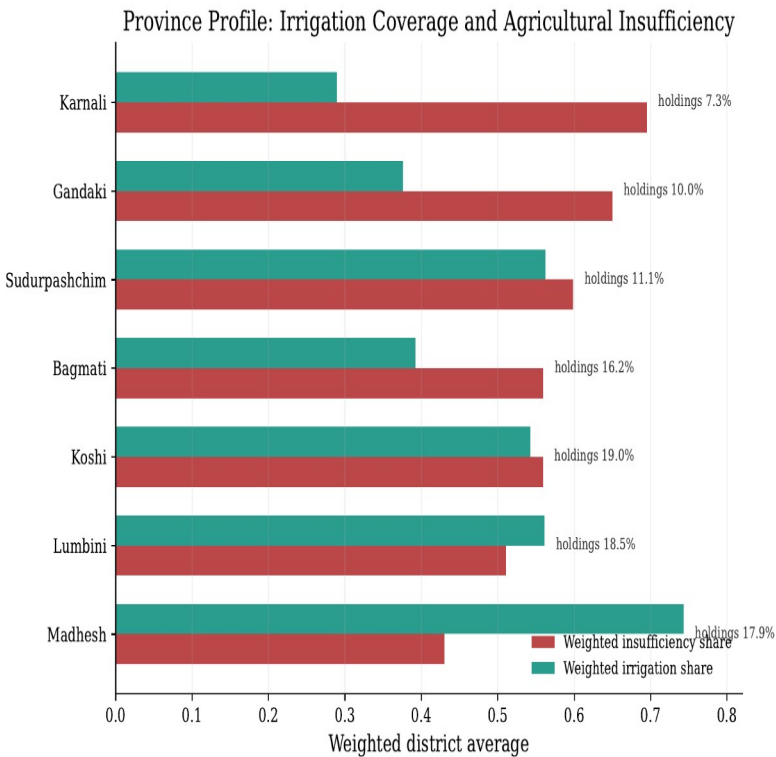
Irrigated-area share

3% (Rukum East) to 91% (Bara)

Parcels per farm

1.5 (Lalitpur) to 6.3 (Humla)

Hardest hit: Karnali & Gandaki provinces. Best off: Madhesh (Terai plains).



Note: Province values are weighted by district agricultural holdings. Madhesh stands out for high irrigation and lower insufficiency, while Karnali shows the opposite pattern.

Province profile: irrigation vs insufficiency

Finding 2 — Irrigation is the strongest signal

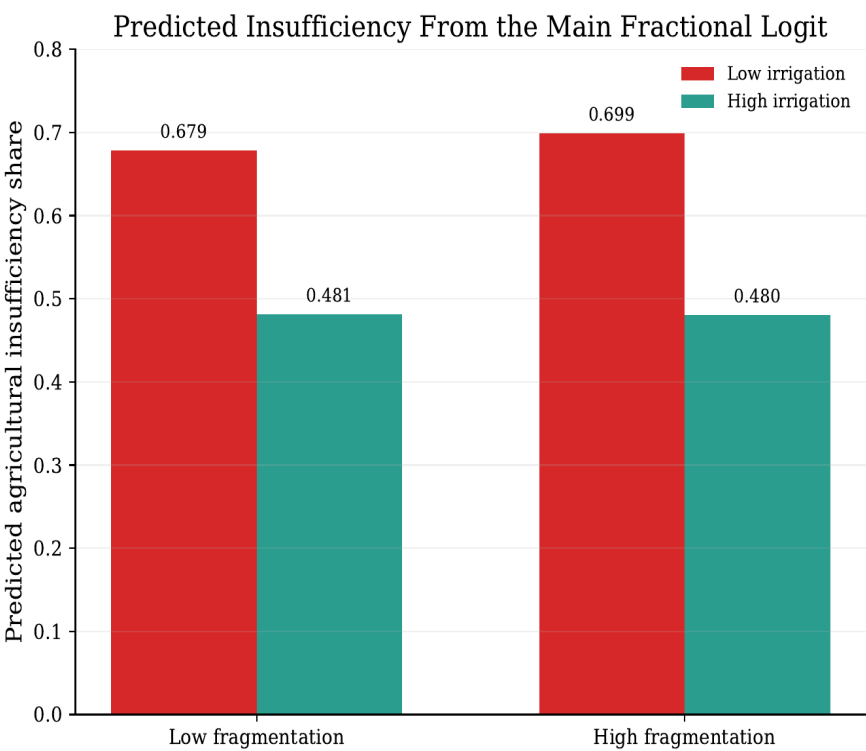
Irrigation is by far the strongest correlate of lower insufficiency (weighted correlation -0.86).

In the model, the predicted insufficiency gap between a low- and a high-irrigation district profile is about:

20 percentage points

lower predicted insufficiency · bootstrap CI excludes zero

This is the study's central, most robust result.



Note: Scenario values come from the main fractional logit using low and high quartile values of fragmentation and irrigation.

Predicted insufficiency: low vs high irrigation

The results, in one table

	Weighted OLS	Fractional logit
Irrigation area share	-0.468***	-1.993***
Avg. parcels (fragmentation)	0.004	0.024
Fragmentation × irrigation	-0.029	-0.193
Six controls + province FE	Yes	Yes
Model fit	R ² = 0.83	fitted r = 0.89
Observations	77	77

Robust HC1 standard errors; holdings-weighted. *** $p < 0.01$.

What the table shows

Only irrigation is significant — at $p < 0.01$ in both estimators. Fragmentation and the interaction are **not significant**.

The average partial effect of irrigation is **-0.45** — the source of the ~20-point scenario gap. The two estimators agree in sign and significance throughout.

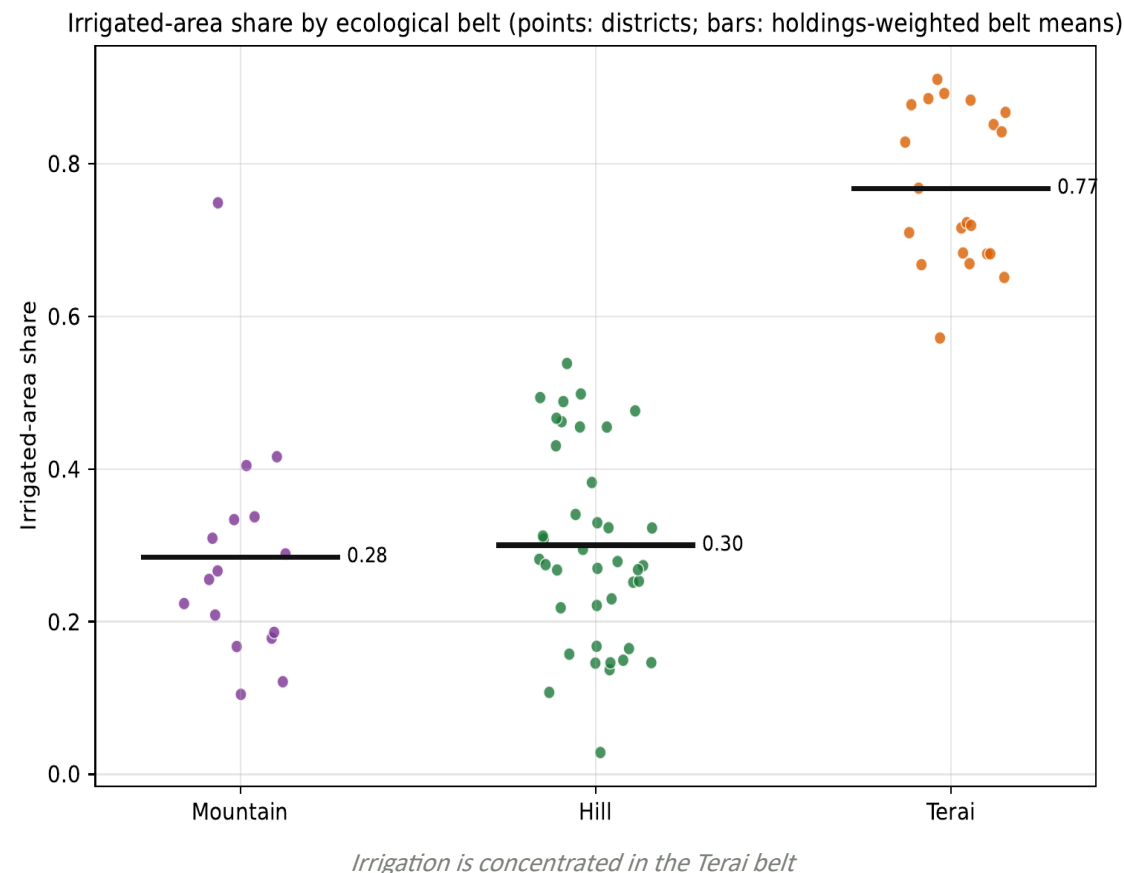
The honest catch — irrigation vs geography

Irrigation and geography are tangled together.

Irrigation sits in the flat **Terai**; insufficiency sits in the **mountains**. When the model controls for the ecological belt, the irrigation effect **shrinks** but stays significant.

Part of the raw irrigation gradient is really geography: the coefficient attenuates from -1.99 (province controls) to -1.30 (belt controls), still significant at $p < 0.01$.

So the honest claim is a careful one: **a strong association we cannot prove is cause.**



Diagnostics — the model holds up

Standard post-estimation checks on the main model:

Concern	Test	Result	Verdict
Heteroskedasticity	Breusch–Pagan	$p = 0.010$	present → HC1 robust SE used
Residual normality	Jarque–Bera	$p = 0.49$	not rejected — residuals normal
Multicollinearity	Maximum VIF	5.39	below 10 (after mean-centring)
Spatial autocorrelation	Moran's I	-0.10 ($p \approx 0.15$)	none — province FE absorb it

Inference is sound: the irrigation result is not an artefact of heteroskedasticity, non-normality, collinearity, or spatial clustering.

Finding 3 — the fragmentation puzzle

On its own

more fragmentation → more insufficiency

≈ **+0.40** bivariate correlation

After controlling for geography

the independent link disappears — not statistically significant

≈ **0** conditional

Why it matters

Fragmented districts are mostly mountain districts. Once we account for that, fragmentation itself adds little. **Lesson: consolidating land alone is unlikely to be a silver bullet for food sufficiency** — a useful, policy-relevant null result (echoing evidence from Rwanda).

RESULTS

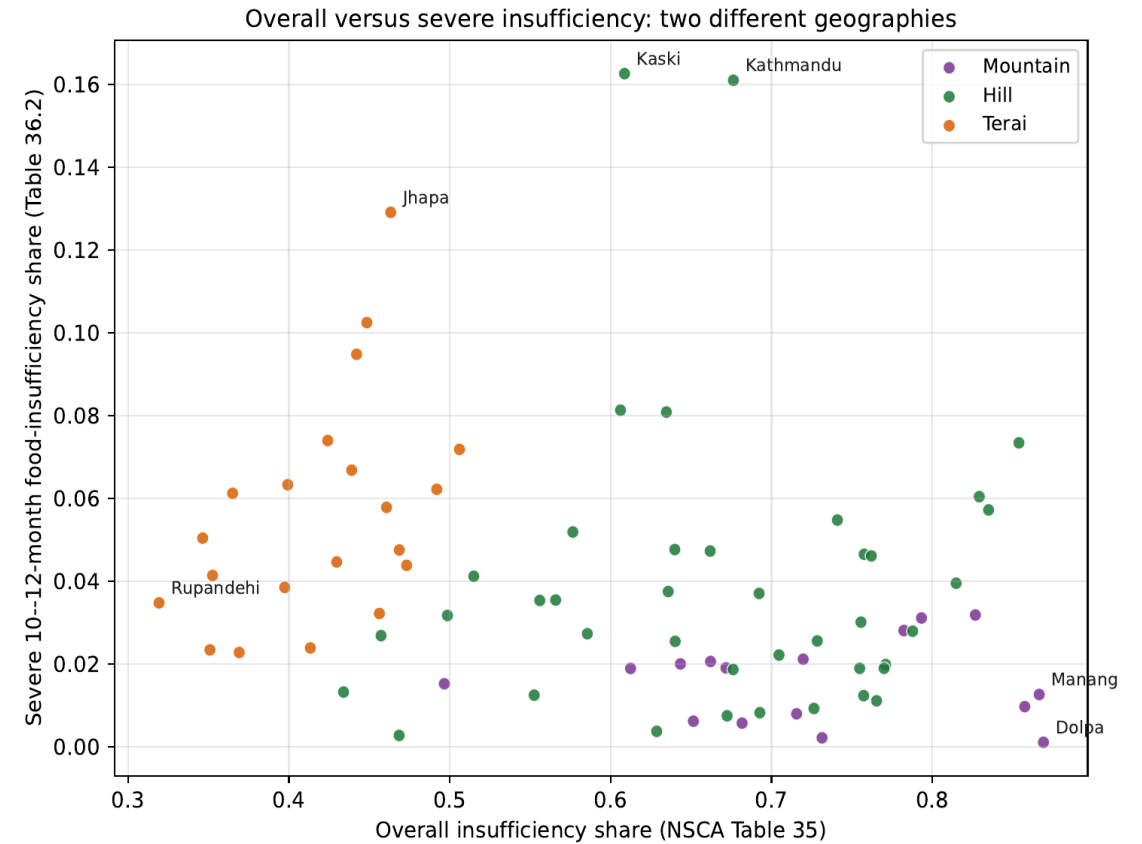
A twist — severe hunger is a different problem

Severe food insufficiency (10–12 months a year) has the **opposite geography**.

It is highest not in the mountains but in **irrigated, urban districts** — Kathmandu, Kaski — where farms are tiny and families live mostly on **off-farm income**.

Lesson

“Can’t grow enough” is not the same as “goes hungry.” The two measures capture different problems and must not be confused.



Overall vs severe insufficiency — barely overlap

Where this sits in the literature

Structure vs. management

The fragmentation null sides with the **“management, not structure”** reading of Nepalese agriculture: once geography and inputs are held fixed, how land is split matters little — what matters is how it is watered and worked.

In international context

Studies from China and Vietnam report fragmentation penalties; this district cross-section finds **none** once geography is controlled — closer to the Rwandan evidence that consolidation gains are modest. Irrigation's dominance echoes the water-centred food-security literature.

What this study adds

The **first nationwide, district-level study** to examine land fragmentation, irrigation and their interaction **together** on Nepal's latest agricultural census, with a fully reproducible workflow.

What it means — for policy

Phrased carefully, because the evidence is associational, not causal:

Prioritise irrigation

Expand reliable, year-round water in low-coverage hill and mountain districts, where the gap is widest.

Bundle, don't isolate

Pursue land consolidation only together with irrigation and machinery — not as a stand-alone fix.

Target chronic hunger differently

Severe deprivation needs help aimed at near-landless, peri-urban households — not only mountain farms.

Ready for local government

District-level evidence (in the annex) can guide provincial and local planning directly.

Honesty about the limits

Stated plainly — these bound the claims, and naming them is part of the rigour:

Association, not cause

The design shows links across districts, not proof that one thing causes another.

Districts, not households

Patterns across districts need not hold for individual farms (ecological inference).

Self-reported outcome

Sufficiency is the farmer's own judgement, against household need.

Small sample

Only 77 districts, so some effects are estimated imprecisely.

One snapshot

A single census year (2021/22); conditions may have shifted since.

A simple fragmentation measure

Counts plots but not how far apart they are.

Contribution and bottom line

First nationwide, district-level study to examine land fragmentation, irrigation and production sufficiency **together**, using the latest census — with a fully reproducible workflow.

THE BOTTOM LINE

Across Nepal's districts, it is water — and the geography it follows — that separates the farms that grow enough to feed their own household from those that fall short. Land fragmentation, on its own, does not.

Substantively: irrigation is the clearest lever. Methodologically: a transparent, reproducible template for district-level research.

Thank you

Questions & discussion